Predicting Heart Disease Risk Using Hybrid Machine Learning

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Section A

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IT Applications

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Introduction

Cardiovascular diseases have been one of the main reasons for fatalities around the world. According to health reports around the globe, heart disease causes millions of fatalities on an annual basis. Therefore, it is important to detect the risk factors of heart disease at an early stage to avoid severe medical complications. Medical experts have always depended on physical examinations and medical tests to detect heart disease. However, it is difficult to detect the risk factors of heart disease since it requires analysing various health factors that influence heart disease.

Machine learning has provided tools that assist in analysing health data and identifying various patterns that may be difficult to identify using basic analysis. Machine learning algorithms assist medical experts in identifying patients who are at risk of suffering from heart disease. This project aims to develop a machine learning algorithm that predicts heart disease risk based on patient health indicators. The project will use a hybrid approach of machine learning that will include supervised and unsupervised learning.

The main problem statement of the project is to use machine learning to detect heart disease and its implications on healthcare.

Dataset Description

The UCI Machine Learning Repository Heart Disease Dataset has been utilised as the dataset in this project. It is a well-known dataset for machine learning studies aimed at predicting cardiovascular disease.

The data set includes medical records of 303 patients and has 13 clinical attributes that address cardiovascular health.

Key features in the dataset include:

Feature	Description
Age	Age of the patient
Sex	Gender of the patient
Chest Pain Type	Type of chest pain experienced
Resting Blood Pressure	Blood pressure at rest
Cholesterol	Serum cholesterol level
Fasting Blood Sugar	Blood sugar level
Resting ECG	Electrocardiogram results
Maximum Heart Rate	Heart rate achieved during exercise
Exercise-Induced Angina	Chest pain triggered by exercise
ST Depression	Depression was observed on the ECG.
Number of Major Vessels	Number of visible major blood vessels
Thalassemia	Blood disorder indicator
Target	Presence of heart disease

The target variable represents whether a patient has heart disease:

- 0 → No heart disease
- 1 → Heart Disease present

The issue may be defined as a binary classification problem because the target variable may have two possible outcomes.

```
import pandas as pd

df = pd.read_csv("heart Disease Data.csv", header=None)
df.head()
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0	6.0	0.0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0	3.0	2.0
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0	7.0	1.0
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0	3.0	0.0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0	3.0	0.0

Sample of the Heart Disease Dataset

Tools and Technologies Used

The implementation of the project was done in the Python programming language in Google Colab, which is an environment to conduct machine learning experiments in the cloud.

There were numerous Python libraries employed to execute the machine learning pipeline and data analysis.

Library	Purpose
Pandas	Data manipulation and dataset loading
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualizations
Scikit-learn	Machine learning algorithms

Google Colab was chosen because it allows easy execution of Python code and provides access to machine learning libraries without requiring local installation.

Data Preprocessing

The dataset was also pre-processed in a number of steps before subjecting it to machine learning algorithms to facilitate the quality of the data and model.

Data Cleaning

The data was analysed to detect missing or invalid data. Entries that have a value of the placehold word like "? were substituted with missing data, and the dataset was excluded to avoid erroneous training of the model.

Data Type Conversion

To make use of machine learning algorithms and to make all variables mathematically calculable, all variables were transformed into numerical values.

Feature Scaling

Standardisation was done through the Standard Scaler, which alters numerical characteristics to take a mean of 0 and a standard deviation of 1.

The scaling of features is especially relevant with those algorithms that use distance estimates between the points of the data, e.g. clustering.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to gain a better insight into the dataset and establish connections between variables.

Distribution of Heart Disease

The results were plotted in a count plot to show the distribution of the number of heart disease and non-heart disease patients. This was used to establish whether the dataset was balanced.

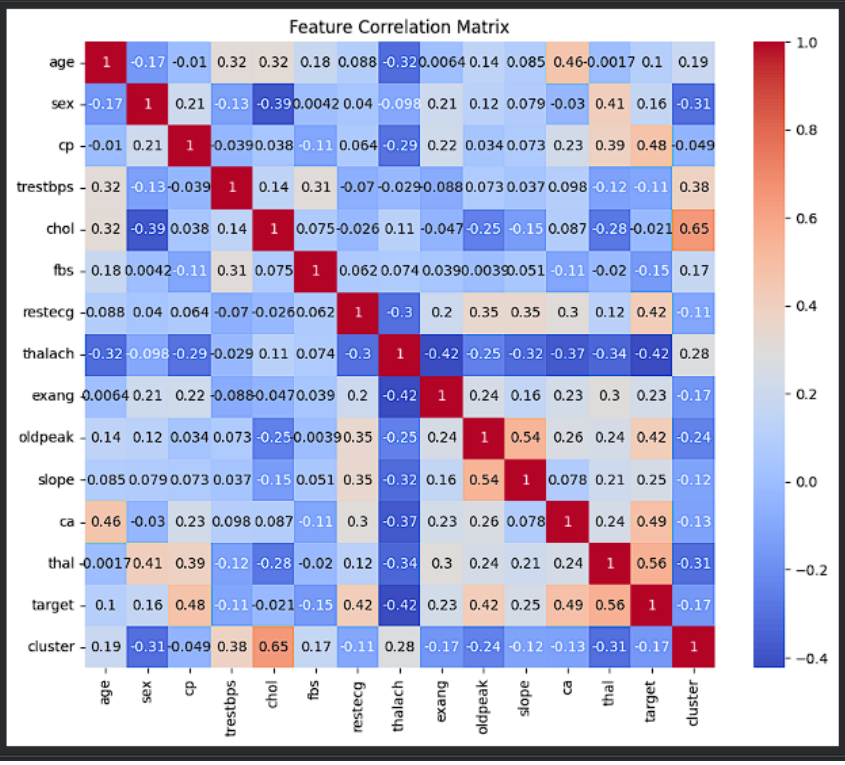
Correlation Analysis

The heatmap of correlation was drawn to see the relationships between various health indicators. The heatmap indicates the variables that have a strong relationship with heart disease.

As an illustration, some of the characteristics, like cholesterol, blood pressure, and evidence of chest pain, were significantly associated with the target variable. The exploratory analysis was used to determine the most meaningful features for the machine learning models.

```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Feature Correlation Matrix")
plt.show()
```



Correlation Matrix of Medical Features

Hybrid Machine Learning Approach

The project uses a hybrid machine learning model that uses two types of learning methods:

Learning Type	Algorithm	Purpose
Unsupervised Learning	K-Means Clustering	Identify patient risk groups
Supervised Learning	Logistic Regression / Random Forest	Predict heart disease risk

The unsupervised learning assists in finding latent patterns in patient records, whereas the supervised learning constructs predictive analytics that categorise patients according to the risk of disease.

This combination allows the system to not only analyse the structure of the dataset but make accurate predictions.

Dataset
↓
Data Cleaning
↓
Feature Scaling
↓
K-Means Clustering
↓
Risk Group Identification
↓
Logistic Regression / Random Forest
↓
Heart Disease Prediction

Hybrid Machine Learning Workflow

Unsupervised Learning: K-Means Clustering

The K-Means clustering algorithm was applied to group patients with similar health characteristics. The clustering algorithm uses similarities between data points to calculate distances.

The Elbow Method was used to find the best number of clusters.

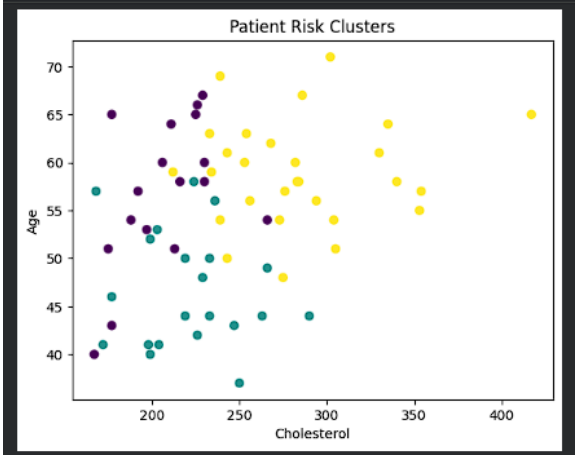
The clustering outcomes generally came up with three groups of patients:

Cluster	Interpretation
Cluster 1	Low-Risk Patients
Cluster 2	Moderate-Risk Patients
Cluster 3	High-Risk Patients

The analysis assists in the process of revealing concealed patterns in the dataset.

```
import matplotlib.pyplot as plt

plt.scatter(df["chol"], df["age"], c=df["cluster"])
plt.xlabel("Cholesterol")
plt.ylabel("Age")
plt.title("Patient Risk Clusters")
plt.show()
```



Patient Clusters Based on Health Indicators

Supervised Learning Models

There were two predictive models that were put into practice to categorize patients.

Logistic Regression

Logistic regression is used to determine the likelihood of heart disease depending on the characteristics of the patient. It is widely applicable in health research since it is not difficult to interpret.

Random Forest

Random Forest is an ensemble algorithm that uses a combination of several decision trees to come up with more precise predictions. It is able to represent the complicated relationships between medical indicators.

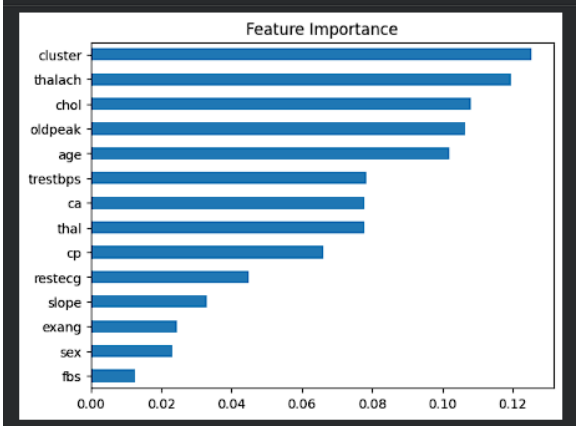
```
import pandas as pd

importance = pd.Series(rf_model.feature_importances_, index=X.columns)

importance.sort_values().plot(kind="barh")

plt.title("Feature Importance")

plt.show()
```



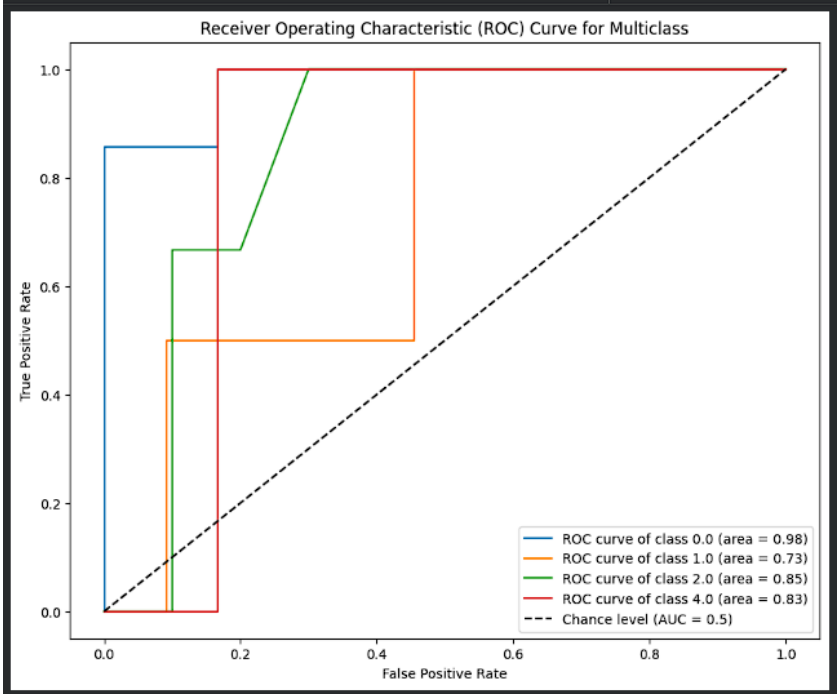
Feature Importance in Heart Disease Prediction

Model Evaluation

The data was split into the Training Set and Testing Set of 80 and 20 percent respectively.

Several performance metrics were used to evaluate the models, such as **Accuracy, Precision, Recall, Confusion Matrix, and ROC Curve.** These measures give a total assessment of the performance of the models

```
plt.plot([0, 1], [0, 1], 'k--', label='Chance level (AUC = 0.5)')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve for Multiclass')
plt.legend(loc='lower right')
plt.show()
```



ROC Curve for Model Performance Evaluation

Conclusion

In this project, the authors show the potential of machine learning to predict the risk of heart disease based on indicators of patient health status. The hybrid machine learning methodology permits pattern finding as well as forecasting modelling, hence applicable in healthcare analytics.

These findings indicate that machine learning models are effective to detect significant cardiovascular risk factors and predict the outcomes of the diseases. These predictive systems can be effective to help in preventive healthcare, clinical decision-making by physicians, and patient outcomes.

References

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). Heart disease [Dataset]. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/45/heart+disease